

JKBOSE 2026

CLASS 11TH

Plant Growth and Development

LECTURE-01

BOTANY

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Topics To Covered

- 1 Introduction ✓
 - 2 Features of growth ✓
 - 3 Phases of growth ✓
 - 4 Growth rates ✓
 - 5 Quantitative comparison of growth
 - 6 Conditions for growth
- Next lecture...
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Introduction

Growth

- It is permanent, irreversible increase in

↗ No. of cells.
↘ Size of cells.
- It is fundamental property i.e. present in all living organisms.
- Growth involves both Catabolism and Anabolism.

Anabolism > Catabolism
(ex. photosynthesis) (ex. Respiration).

- Growth involves increase in protoplasm of cell (Internal / Intrinsic growth).
Ex. Expansion of leaf is growth.

NOTE 1. Initial stage of plant growth is seed germination.

2. Seed germination is of two types:

a. Epigeal Seed Germination: Cotyledons of seed come out of soil during germination.

b. Hypogeal Seed Germination: Cotyledons of seed does not come out of soil during germination.

Features of Growth



1. Growth is **Conspicuous** i.e, visible.
2. Plant Growth is Unlimited (**indeterminate**) i.e, Continuous growth because of presence of meristematic tissue.

Exception: Some parts of plant eg. leaf, fruit, flower show determinate growth (limited growth)

3. Plant growth is **open type** i.e, new cells are being continuously added due to presence of meristematic tissue.

4. Plant growth is **localised** i.e, occurs at fixed positions where meristematic tissues are present.

Ex Apical meristem (RAM, SAM) $\rightarrow$ Cause Increase in length (1° growth) of root and stem.

Lateral meristem (vascular Cambium) $\rightarrow$ Cause Increase in girth (2° growth) of plant.

5. Plant growth is **quantitative** i.e., measurable.

≡ Growth is actually increase in protoplasm of cell which is difficult to measure directly.

≡ By some indirect methods, plant growth can be measured:

a.	Increase in	fresh weight	} Example is leaf.
b.	"	dry weight	
c.	"	Surface area	

d. Increase in length ex. pollen tube, Root, Stem etc.

e. Increase in volume ex. Volume of watermelon increases 3,50,000 times

f. Increase in number of cells ex. RAM of single maize grain produces 17,500 new cells per hour.

Phases of Growth

≡ 03 Phases/Zones of Growth →

1. Meristematic phase/zone.
2. Elongation phase/zone.
3. Maturation phase/zone

Already Discussed
in Morphology chapter



Growth Rates



≡ Growth rate is defined as "growth per unit time"

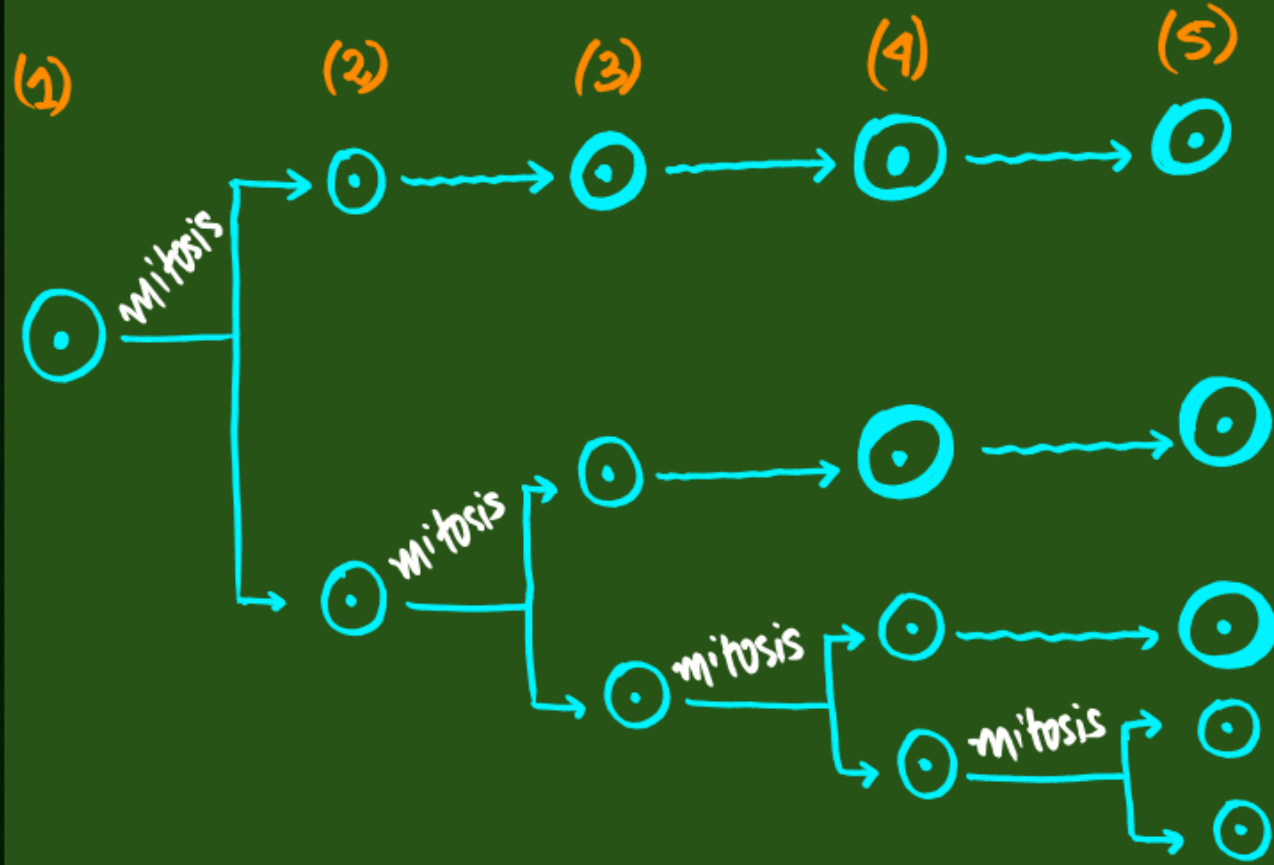
$$\text{Growth rate} = \frac{\text{Growth}}{\text{Unit time}}$$

≡ Growth rate – 02 types:

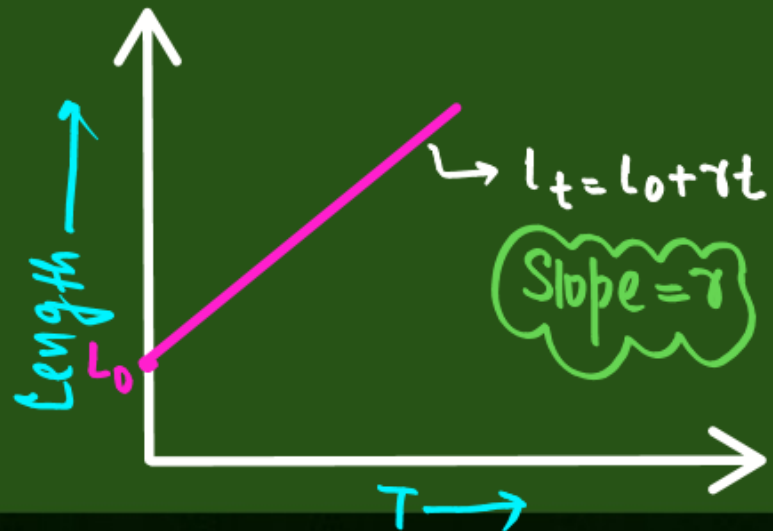
a. Arithmetic growth rate.

b. Geometric growth rate.

Arithmetic Growth rate



Graph of Arithmetic Growth rate:



$\Rightarrow$ Here a cell divides (by mitosis) to form two cells — out of which one cell becomes permanent and 2nd cell continues to divide

$\Rightarrow$ Increase in number of cells is constant w.r.t time. (1, 2, 3, 4, 5, 6, ...)

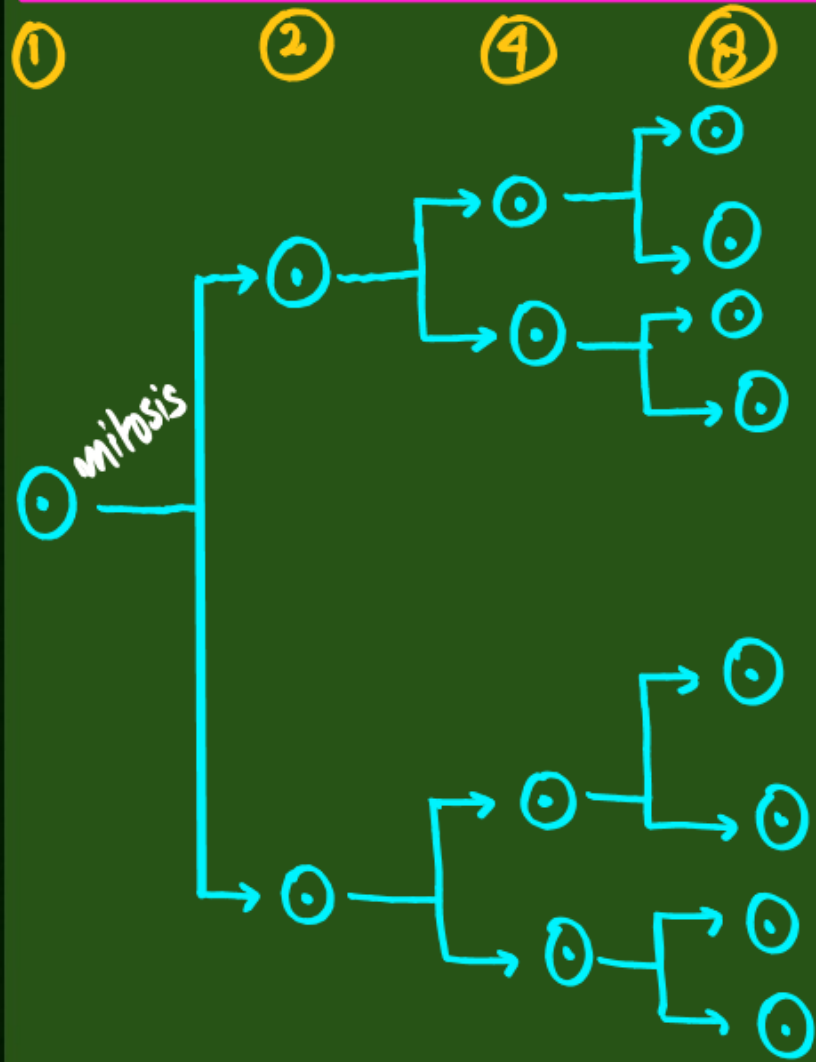
Equation of Arithmetic Growth rate:

$$L_t = L_0 + \gamma t$$

L_t = Final length
 L_0 = Initial length
 γ = Arithmetic growth rate
 t = time period.

It represents straight line equation ($y = mx + c$)

Geometric Growth Rate



- ≡ Here number of cells doubles after every division
- ≡ One cell divides mitotically to form two cells and then both cells continuously divide.
- ≡ Here no. of cells increases geometrically (1, 2, 4, 8, 16, 32, 64, ...)

Equation of Geometric Growth rate:

$$W_t = W_0 e^{rt}$$

↓
S-shaped (Sigmoid) Graph.

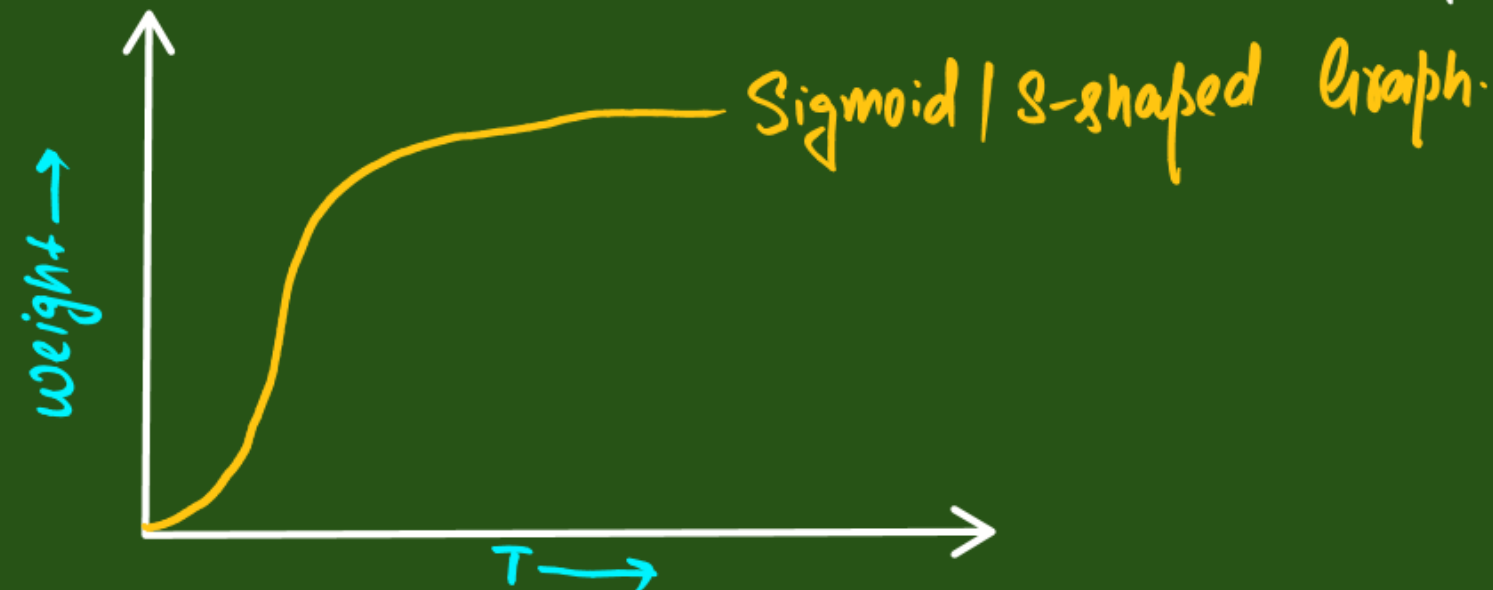
W_t = Final weight

W_0 = Initial "

e = base of natural logarithm.

r = Geometric growth rate.

t = time period.



Quantitative Comparison of Growth



Next lecture

Conditions of Growth



Next lecture

HOMEWORK

Revision





Thank
You